Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)			layers/sheets of hexagons of C atoms bonded together (1) weak forces between layers (allowing layers to move and making it soft) (1)	2			2		
	(b)	(i)		121 kJ mol^{-1}	1			1		
		(ii)		enthalpy of atomisation for Cl is 121 (1) correctly constructed energy cycle or expression e.g. $\Delta_f H(NaCl) = \Delta_{at}H(Na) + IE(Na) + ½BE(Cl_2) + EA(Cl) + \Delta_{latt}H(NaCl)$ (1) $\Delta_{latt}H(NaCl) = –771$ kJ mol^{-1} (1)		3		3	2	
		(iii)		student is incorrect must consider entropy of surroundings as well / must consider effects of enthalpy on entropy of surroundings / Gibbs free energy must be considered (must be negative and this includes enthalpy and entropy) (1) award (1) for either of following entropy change for this reaction will be negative as gas is removed entropy change will be negative as entropy of chlorine / gaseous reactant is greater than entropy of product		1	1	2		

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Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
	(c)			**Indicative content** • brick red flame – one of the metals must be calcium • no other colour – other metal may be magnesium / cannot be lithium/sodium/strontium/barium (ignore references to potassium unless clearly indicated that colour is weak and could be hidden by colour due to calcium) • cloudiness with dilute sulfuric acid due to calcium sulfate being sparingly soluble (but no precipitate so no strontium/barium present) • precipitate with silver nitrate – X must be chloride/bromide/iodide • misty fumes with sulfuric acid are hydrogen halides • no coloured fumes so must be chloride • M_r of water is 216.27 so d = $\frac{216.27}{18.02}$ = 12 • M_r of anhydrous solid is 301.73 • formula must be $CaMg_2Cl_6.12H_2O$ (allow without H_2O as long as $12H_2O$ has been clearly calculated earlier) **5-6 marks** The candidate includes at least six relevant points and correctly identifies the formula *The candidate constructs a relevant, coherent and logically structured account including key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary is used accurately throughout.* **3-4 marks** The candidate includes at least four relevant points and correctly identifies all ions present *The candidate constructs a coherent account including many of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary is generally sound.* **1-2 marks** The candidate includes at least three relevant points *The candidate attempts to link relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material. There is some evidence of appropriate use of scientific conventions and vocabulary.* **0 marks** *The candidate does not make any attempt or give an answer worthy of credit.*	2	2	2	6	1	5
				Question 8 total	**5**	**6**	**3**	**14**	**3**	**5**

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